



University
of Exeter

COM3031 Probabilistic Machine Learning Week 4

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In this week

Lectures

- Variational Inference
- Kullback–Leibler (KL) Divergence
- Variational Lower Bound and Optimising THE ELBO

Workshop

- Bayesian Linear Regression and GLM

Acknowledgements: some lecture slides were adapted from Wesley Maddox, Michael Gutmann, Willie Neiswanger and Dmitry Vetrov.

Part I: Variational Inference

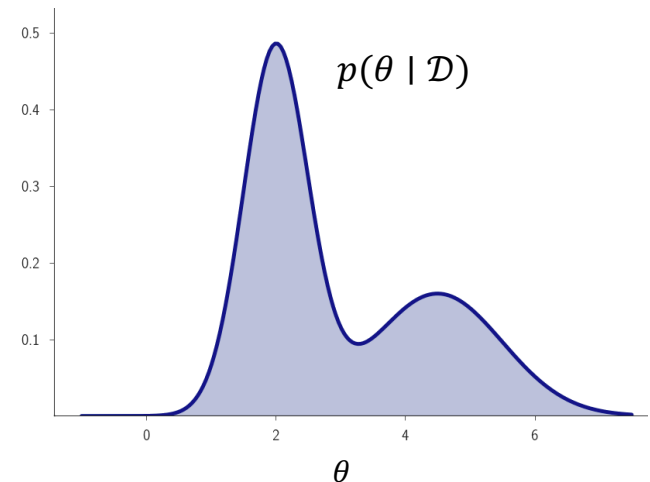
Approximation Bayesian Inference

In Bayesian inference, we aim to compute the posterior $p(\theta \mid \mathcal{D})$

- Exact posteriors are available only for a few simple conjugate models (e.g. Gaussian–Gaussian).
- In most realistic models, the required integrals are intractable.

Posteriors often look like the figure, we replace $p(\theta \mid \mathcal{D})$ with $q(\theta)$

- **Laplace approximation:** Local Gaussian approximation
- **Variational inference:** find the “best” approximating distribution via optimisation
- **Markov Chain Monte Carlo (MCMC):** use samples to represent the posterior



An example of a realistic complex (non-gaussian, skewed and multimodal) posterior distribution

Why Laplace Approximation Is Not Enough

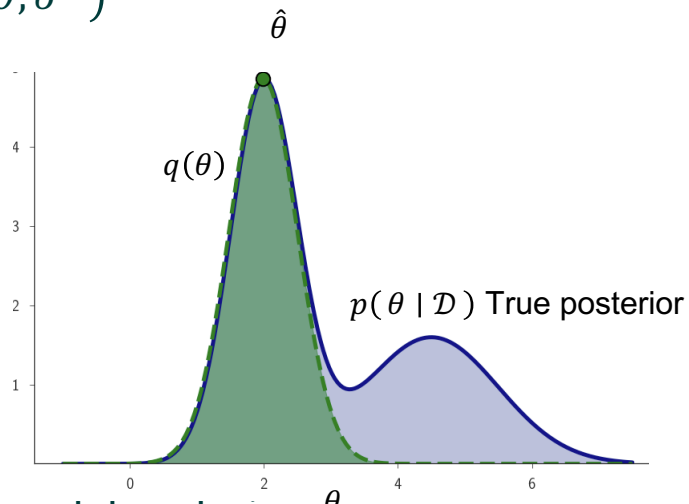
The Laplace approximation **finds a Gaussian approximation** $q(\theta)$ which is centred on the mode of the posterior distribution $p(\theta)$.

$$p(\theta | \mathcal{D}) \approx q(\theta) = \mathcal{N}(\theta | \hat{\theta}, \sigma^2)$$

- $\hat{\theta}$ = **MAP estimate** (posterior mode)
- σ^2 : inverse local curvature at the mode

Key limitations

- Local only and Captures only one peak
- Uses information *only near the mode*
- Assumes near-Gaussian shape
- Forces symmetry Fails for skewed posteriors and multimodal posteriors



Variational Inference

Goal: approximate an intractable posterior with a simpler distribution.

$$p(\theta | \mathcal{D}) \approx q(\theta)$$

Key idea (vs Laplace):

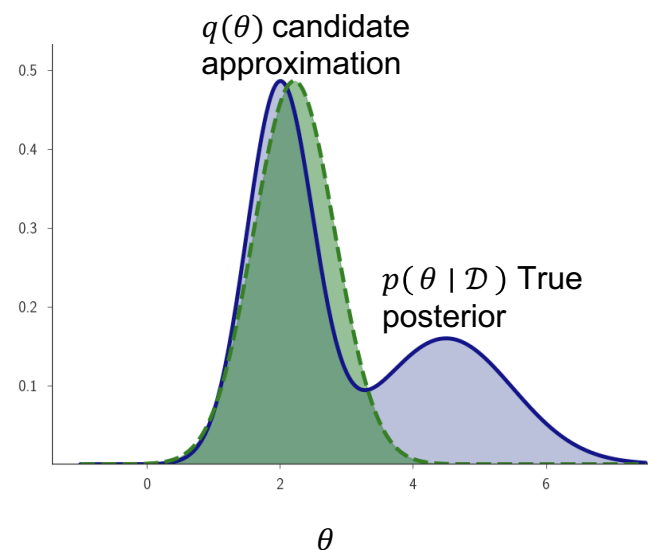
- Laplace: local Gaussian near the MAP
- VI: choose $q(\theta)$ to be close to $p(\theta | \mathcal{D})$ globally

How we do it:

- Choose q from a family of distributions (e.g. Gaussians, factorised forms) $q(\theta) \in \mathcal{Q}$
- Then solve an optimisation problem

$$q^* = \arg \min_{q \in \mathcal{Q}} \text{distance}(q(\theta), p(\theta | \mathcal{D}))$$

- Question: what distance should we use?



Choosing the Distance: KL Divergence

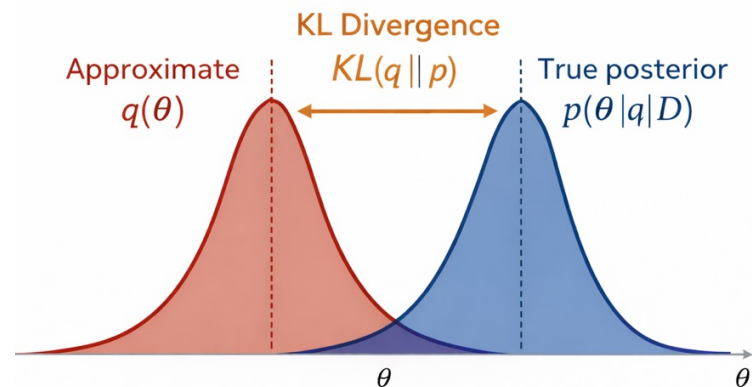
- **Standard choice in VI:** minimise the KL divergence

$$\text{KL} (q(\theta) || p(\theta | \mathcal{D})) = \int q(\theta) \log \frac{q(\theta)}{p(\theta | \mathcal{D})} d\theta$$

- **Variational inference objective:**

$$q^* = \arg \min_q \text{KL} (q(\theta) || p(\theta | \mathcal{D}))$$

- **Interpretation:** we choose the q that is “closest” to the posterior within the family we allowed.



Part II: Kullback–Leibler (KL) Divergence

KL Divergence: Measuring Approximation Error

- **Goal:** measure how close an approximation $q(x)$ a target distribution $p(x)$
- The **Kullback–Leibler (KL) divergence** from q to p :

$$\text{KL}(q \parallel p) = \int q(x) \log \frac{q(x)}{p(x)} dx$$

Interpretation:

- KL measures the **expected log difference** between p and q
- It is an **expectation** under the approximation q :

$$\text{KL}(q \parallel p) = \int q(x) (\log q(x) - \log p(x)) dx = E_q[\log q(x) - \log p(x)]$$

- Intuitively, how much information is lost when we use q instead of p ?

Key properties:

- **Non-negative:** $\text{KL}(q \parallel p) \geq 0$ with equality iff $q(x) = p(x)$
- **Asymmetric:** $\text{KL}(q \parallel p) \neq \text{KL}(p \parallel q)$, the direction matters

Mathematical intuition on KL Divergence

Idea: The non-negativity of KL divergence follows from convexity and Jensen's inequality.

Convexity reminder:

- A function f is convex if:

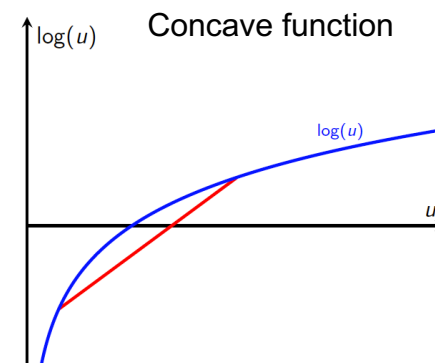
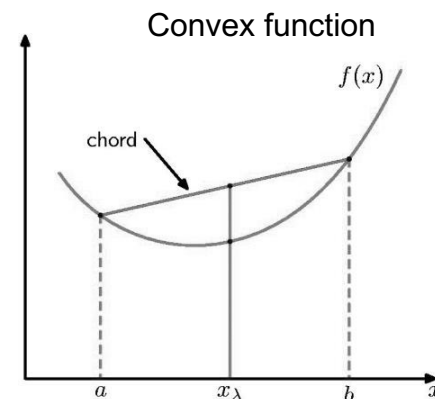
$$f(\lambda a + (1 - \lambda)b) \leq \lambda f(a) + (1 - \lambda)f(b)$$

- The function $-\log(x)$ is convex, while $\log(x)$ is concave
- Jensen's inequality

$$f\left(\sum_{i=1}^M \lambda_i x_i\right) \leq \sum_{i=1}^M \lambda_i f(x_i) \text{ where } \lambda_i \geq 0 \text{ and } \sum_i \lambda_i = 1$$

Apply to KL divergence:

$$\begin{aligned} \text{KL}(q|p) &= E_q \left[\log \frac{q(\theta)}{p(\theta)} \right] \\ &= E_q[-\log p(\theta)] - E_q[-\log q(\theta)] \geq 0 \end{aligned}$$



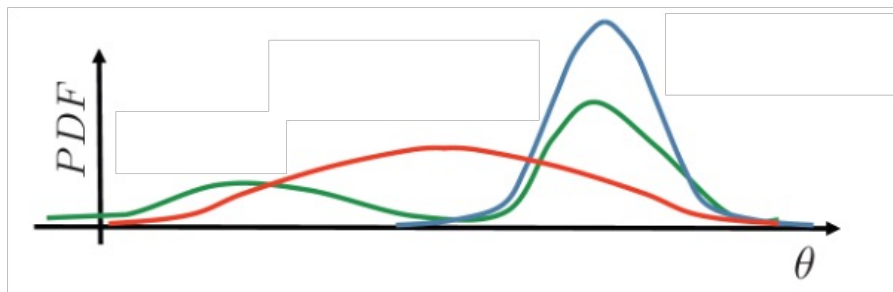
KL Divergence

Reverse KL: (“zero forcing”, mode seeking)

- We choose q by minimising:

$$q^* = \operatorname{argmin}_q KL(q||p) = \operatorname{argmin}_q \int q(x) \log \frac{q(x)}{p(x)} dx$$

- The expectation is taken **under** q .
- If $q(x) > 0$ where $p(x) \approx 0$, then $\log \frac{q(x)}{p(x)}$ becomes very large $\rightarrow$ big penalty.
- So the optimiser tends to place q only in regions where p is large.
- Produces good local fit, “mode seeking”



Green: a bimodal distribution $p(x)$
Blue: minimising $KL(q||p)$
Red: minimising $KL(p||q)$

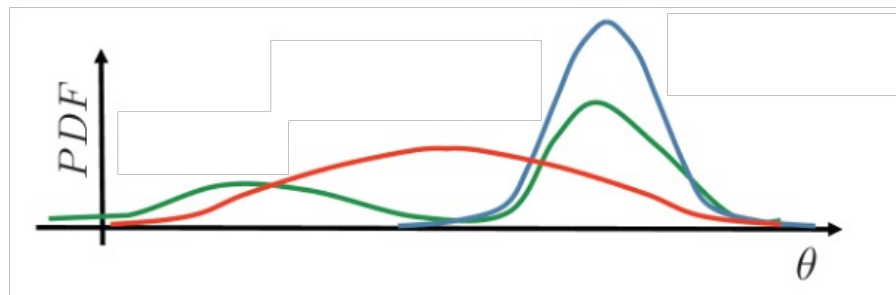
KL Divergence

Forward KL: (zero avoiding)

- We choose q by minimising:

$$q^* = \operatorname{argmin}_q KL(p||q) = \operatorname{argmin}_q \int p(x) \log \frac{p(x)}{q(x)} dx$$

- Forward KL penalises heavily when $q(x)$ is small where $p(x)$ is large
- As a result, the optimal q tries to **cover all regions where p has mass**
- Corresponds to MLE; produces global fit/moment matching



Green: a bimodal distribution $p(x)$
Blue: minimises $KL(q||p)$
Red: minimises $KL(p||q)$

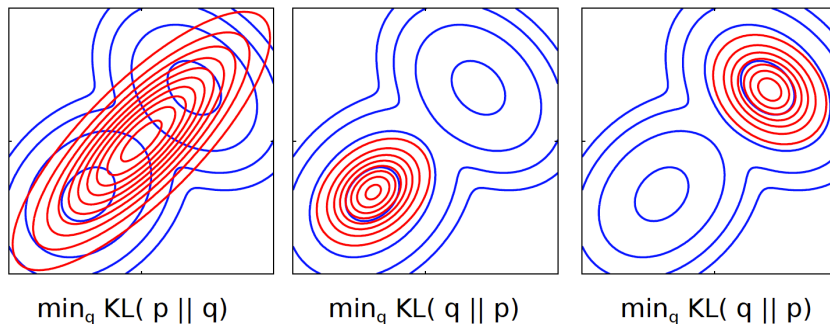
KL Divergence: Forward vs Reverse

Forward KL:

- q is pushed to **cover all regions** where p has probability mass
- Missing any mode of p is heavily penalised
- Result: a **broad approximation** that may overestimate uncertainty

Reverse KL:

- q avoids regions where p is small
- It is cheaper to **focus on one mode** than to cover all modes
- Result: a mode-seeking approximation that underestimates uncertainty



Blue contours: true posterior distribution p .
Red contours: unimodal approximation q

KL Divergence in Variational Inference

Variational inference as optimisation:

$$q^* = \arg \min_q \text{KL} (q(\theta) || p(\theta | \mathcal{D}))$$
$$\text{KL} (q(\theta) || p(\theta | \mathcal{D})) = \int q(\theta) \log \frac{q(\theta)}{p(\theta | \mathcal{D})} d\theta$$

Intuition:

- KL measures how much information is lost when using q instead of p .
- It is an expectation under q .

Why Variational Inference Uses Reverse KL:

- **Reverse KL** chooses q because it is computationally tractable and leads directly to the ELBO, which we can optimize using standard tools.
- **Forward KL** is impractical because it depends on the true posterior p , which we do not know and cannot evaluate.

Part III: Variational Evidence Lower Bound and Optimising the ELBO

Evidence lower bound (ELBO)

Goal in Variational Inference:

$$\text{KL} (q(\theta) || p(\theta | \mathcal{D})) = \int q(\theta) \log \frac{q(\theta)}{p(\theta | \mathcal{D})} d\theta = E_q [\log q(\theta) - \log p(\theta | \mathcal{D})]$$

Problem:

- The true posterior $p(\theta | \mathcal{D})$ is intractable, so the KL divergence cannot be computed directly.

From KL to ELBO:

- Using Bayes' rule,

$$p(\theta | \mathcal{D}) = \frac{p(\mathcal{D} | \theta)p(\theta)}{p(\mathcal{D})}$$

- Substitute into the KL:

$$\text{KL} (q(\theta) || p(\theta | \mathcal{D})) = E_q [\log q(\theta)] - E_q [\log p(\mathcal{D} | \theta)] - E_q [\log p(\theta)] + \log p(\mathcal{D})]$$

- Rearranging terms,

$$\log p(\mathcal{D}) = E_q [\log p(\mathcal{D} | \theta)] - \text{KL}(q(\theta) || p(\theta)) + \text{KL} (q(\theta) || p(\theta | \mathcal{D}))$$

Evidence lower bound (ELBO)

Evidence Lower Bound (ELBO):

$$\mathcal{L}(q) = E_q[\log p(\mathcal{D} | \theta)] - \text{KL}(q(\theta) || p(\theta))$$

- Depends only on the likelihood $p(\mathcal{D} | \theta)$, the prior $p(\theta)$ the variational distribution $q(\theta)$.
- All expectations are taken under $q \rightarrow$ tractable.
- Turns Bayesian inference into a standard optimisation problem.

KL divergence is always nonnegative,

$$\log p(\mathcal{D}) = \mathcal{L}(q) + \text{KL}(q(\theta) || p(\theta | \mathcal{D}))$$

If $q = p(\theta | \mathcal{D})$, the gap is zero, otherwise $\mathcal{L}(q) \leq \log p(\mathcal{D})$

- **Full derivation of ELBO is available on ELE.**

Optimising ELBO

Maximising ELBO $\Leftrightarrow$ Minimising KL divergence

$$\text{KL} (q(\theta) || p(\theta | \mathcal{D})) = -\mathcal{L}(q) + \log p(\mathcal{D})$$

Since $\log p(\mathcal{D})$ is constant with respect to θ ,

$$\arg \min_q \text{KL} (q(\theta) || p(\theta | \mathcal{D})) \Leftrightarrow \arg \max_q \mathcal{L}(q)$$

Problem: we cannot optimise over *all* possible distributions.

Instead, we restrict $q(\theta)$ to a simpler family $\mathcal{Q}$. This gives two common strategies:

- **Mean-field variational inference**

Assume $q(\theta)$ *factorises* into independent components

- **Parametric variational inference**

Assume $q(\theta)$ has a fixed parametric form (e.g. Gaussian with learnable parameters)

Mean-Field Variational Inference

Motivation: make optimisation tractable

- Variational inference requires optimising over distributions, which is hard in general.
- To make this feasible, we restrict the variational family Q to a simpler form.

Mean-field assumption:

- The simplest restriction: assume latent variables are independent under q .
- Let the latent variables be: $\theta = (\theta_1, \theta_2, \dots, \theta_K)$

Mean-field factorisation: $q(\theta) = \prod_{k=1}^K q_k(\theta_k)$

- Each factor $q_k(\theta_k)$ is a separate variational distribution
- Dependencies in the true posterior are ignored

Mean-Field Variational Inference

Goal: We want to maximise the ELBO over a mean-field family:

$$q^* = \operatorname{argmax}_{q_1, \dots, q_K} \mathcal{L}(q) \quad \text{subject to } q(\theta) = \prod_{k=1}^K q_k(\theta_k)$$

Block coordinate ascent idea: instead of optimising all q_k at once, we update one factor at a time.

$$q_k \leftarrow \operatorname{arg max}_{q_k} \mathcal{L}(q_1, \dots, q_k, \dots, q_K)$$

Key update rule:

- Fix all factors except q_k and define $q_{-k} = \prod_{j \neq k} q_j(\theta_j)$
- Then the maximiser of the ELBO with respect to q_k is:

$$\log q_k^*(\theta_k) = E_{q_{-k}}[\log p(\mathcal{D}, \theta)] + \text{const}$$

- Exponentiating:

$$q_k^*(\theta_k) \propto \exp(E_{q_{-k}}[\log p(\mathcal{D}, \theta)])$$

When do we get “closed-form” updates

What does “closed form” mean here?

- A *closed-form update* means the optimal factor $q_k^*(\theta_k)$ has a known distribution form (e.g. Gaussian, Gamma, Dirichlet),
- so we can update it analytically, without numerical optimisation.

When does this happen?

- In conjugate / exponential-family models, the conditional $p(\theta_k \mid \theta_{-k}, \mathcal{D})$ belongs to a known family.
- Then the update is analytic, $q_k^*(\theta_k) \propto \exp(E_{q_{-k}}[\log p(\mathcal{D}, \theta)])$

Otherwise (the common case)

- If the conditional is not in a convenient family (e.g., logistic regression, neural nets), we can’t write a neat update.
- We use numerical optimisation instead (e.g., gradient ascent on ELBO, stochastic VI).

Block Coordinate Ascent Algorithm

Algorithm:

Initialise: $q(\theta) = \prod_{k=1}^K q_k(\theta_k)$

Repeat until convergence (ELBO stops increasing):

Update one factor at a time

- For each $k = 1, \dots, K$, update q_k while keeping all other factors fixed:

$$q_k^*(\theta_k) \propto \exp(E_{q_{-k}}[\log p(\mathcal{D}, \theta)])$$

- Compute ELBO to monitor progress

Practical note (when is it “closed form”?)

- In conjugate / exponential-family models, q_k^* often has a known distribution form (easy update).
- Otherwise, we use numerical optimisation (e.g. gradients / stochastic VI).

Parametric Variational Approximation

Why move beyond mean-field?

- Mean-field VI assumes: $q(\theta) = \prod_{k=1}^K q_k(\theta_k)$. This gives simple updates, but:
 - strong independence assumptions
 - poor approximations for correlated parameters
 - closed-form updates only exist in conjugate models

Key idea: parameterise the approximation

- Instead of optimising over *all possible distributions*, we restrict q to a **parametric family**:

$$q(\theta) = q(\theta|\lambda)$$

- λ denotes variational parameters (e.g. mean, variance).
 - The shape of q is fixed, but its location and spread adapt
- This turns VI into a **standard optimisation problem**.

Parametric Variational Approximation

Common choices for $q(\theta|\lambda)$

Gaussian family (most common):

$$q(\theta | \lambda) = \mathcal{N}(\theta | \mu, \Sigma) \quad \text{where } \lambda = (\mu, \Sigma)$$

- Can capture **correlations** between parameters.

Diagonal Gaussian (scalable):

$$q(\theta | \lambda) = \mathcal{N}(\theta | \mu, \text{diag}(\sigma^2))$$

- Fewer parameters, faster optimisation, common in deep learning.

Optimisation objective (ELBO)

- We now maximise the ELBO with respect to λ :
- Same ELBO as before, different way of optimising it.
- Use gradient-based optimisation: Gradient ascent and Stochastic gradients (mini-batches)

Parametric variational inference trades analytic simplicity for flexibility and scalability

Summary

Variational inference (VI):

- Approximates the posterior with a chosen distribution q
- Turns inference into an optimisation problem
- Objective: minimise reverse KL

Evidence Lower Bound (ELBO):

- A computable lower bound on the log evidence
- Maximising ELBO $\equiv$ minimising KL divergence

Mean-field VI:

- Assumes independence under q :
- Enables **block coordinate ascent** updates
- Fast and scalable, Often analytic updates

Parametric VI:

- Parameterise q directly (e.g. Gaussian)
- Optimise ELBO using gradients
- More flexible, widely used in modern ML